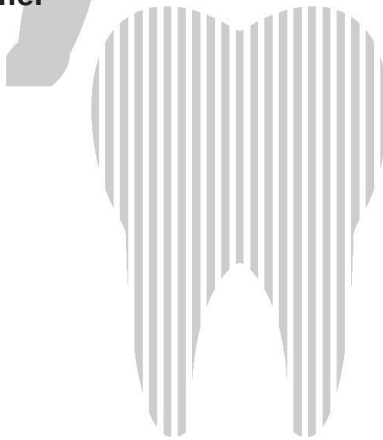


Dental anatomy

"The road to success and the road to failure are almost exactly the same."

Canines



LM
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Canines

General characteristics:

*There are one canine in each quadrant. Canine may be named as Cuspid .

*Canines have longer and stronger roots than those of other teeth .

*over the labial aspect of their roots . there is well developed bone ridge called the canine eminence , giving the canine a very important cosmetic value because they ensure facial expression.

*Canines are formed of 4 lobes (3 labial and 1 lingual) , the middle lobe is highly developed labially forming the labial ridge and lingually forming the lingual ridge and incisally forming a cusp , therefore canines is called the cuspids.

*the canines are located between incisors and premolars, so they support the two groups in function (tearing and partial cutting).

	Begainig of calcification	Crown completion	Eruption	Root completion
$\frac{1}{1}$	3-4 month	3 years before eruption date	7 years	3 years after eruption date
			6 years	
$\frac{2}{2}$	10-12 months		8 years	
	3-4 months		7 years	
$\frac{3}{3}$	4-5 months		11 years	
			9 years	

1) Labial aspect	Upper Canine	Lower Canine
GEOL	Trapezoid with the smallest uneven side cervically and the largest side incisally. They have narrow neck giving the crown fan shaped appearance.  	
All outlines	-Mesial: convex, from cervical line to contact areas -Distal: concave, from cervical line to contact areas -Incisal: Mesial cusp slope : concave Distal cusp slope : convex DCS > MCS -Cervical line : convex root wise	The same but with small differences :- -its width is narrower than the upper canine. -it's crown appear longer than upper canine because of the narrow neck and the more incisally located incisors. -the mesial outline is straight from the cervical line to the contact areas. -the cusp tip is less pointed
Contact areas	MCA: at the junction of middle and incisal 1/3 DCA: more cervically, located in middle 1/3	MCA: in the incisal 1/3 DCA: more cervically. at the junction of middle and incisal 1/3
Surface anatomy	The same but less developed.	
1)Elevation	-The labial surface is convex with the maximum convexity at the cervical 1/3 (cervical ridge). -Labial ridge: prominent from cusp tip to cervical ridge.	
2)Depressions	-2 shallow developmental grooves which divide the labial surface into 3 lobes .	
ROOT	Single root The apex is pointed and <u>curved distally</u> . The root is the longest in all teeth.	Single root curved distally, but slightly shorter than the lower canine.

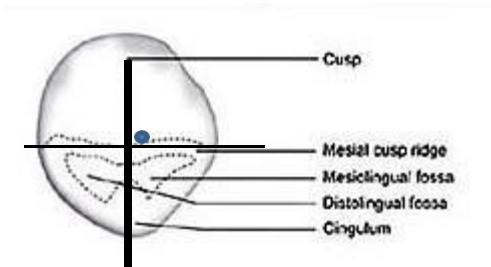
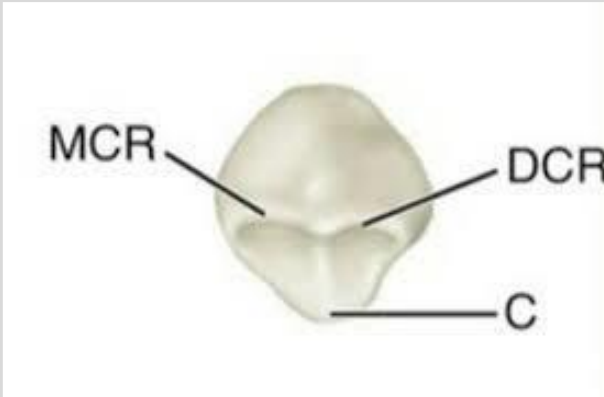
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2)Lingual aspect	Upper Canine	Lower Canine
N.B	Similar to that of the labial surface , but with different size and surface anatomy. Both crown and root are narrower linguallly than labially due to (lingual convergence)  	
Surface anatomy	-Cingulum : is <u>well</u> developed -Mesial and distal marginal ridges : <u>well</u> developed . -The lingual ridge is prominent and extends from cusp tip to cingulum	
1)Elevation	-Cingulum: less developed -Mesial and distal marginal ridges : less developed , -The lingual ridge is restricted and found only in incisal 1/3.	
2)Depressions	- 2 lingual fossae	
	- 1 lingual fossa , due to the less developed lingual ridge.	
Root	The lingual surface is smooth and convex . It's narrower linguallly than labially	

Dr. M.

3) Mesial aspect	Upper Canine	Lower Canine
GEOL	Triangular in shape , the base towards the cervical line and the apex towards the incisal edge .	
The outline form	-Labial outline : convex in the cervical third (cervical ridge) And is straight in the middle and the incisal thirds . Lingual outline : convex in the cervical third (cingulum) then straight in the middle (lingual ridge) and sl. Convex in the incisal ridge . Cervical line : curves incisally to a marked degree in mesial and less curved in distal.	Same as the lower canine but with some differences : -the convexities of the labial outline is less prominent due to less developed anatomical landmarks. - the lingual outline in the middle third is concave due to less prominent lingual ridge.
N.B	The cusp tip of the crown is labial to the center of the root.	-the cusp tip is located lingual to the center of the root due to lingual inclination.
Contact areas	Centralized (labio-lingual) and in the junction between incisal and middle third near	the contact area is centralized and in the incisal third.
Root	-the single root is conical, it's apex is blunt and curved labially. - there is shallow longitudinal depression.	. The developmental depression on the root is more pronounced than that of upper canine

4) distal aspect	Upper canine	Lower Canine
N.B	Same as mesial aspect except :- -The cervical line has less curvature toward the cusp tip -The contact areas is still centralized but is located more cervically. -The depression on the distal aspect of the root is more pronounced than on mesial aspect . - this depression on the root helps in anchorage and stanilization of the tooth in the arch , to prevent rotation and displacement.	
اوعي تنسي دي		The developmental depression on the root in lower canine may be deep that it can divide the root into 2 roots : 1 labial + 1 lingual roots 

4)Incisal aspect	Upper Canine	Lower Canine
GEOL	Diamond , and the crown converges lingually (lingual convergence)	
	Thickness > width	
N.B	-the labial and the lingual outlines is convex more than incisors -the cusp tip is labially and mesially situated (Labially due to : this teeth is upper tooth Mesially due to : the DCS is > MCS) -The labial ridge is very noticeable. -The lingual ridge extends from cusp tip to the cingulum in between the 2 lingual fossae. -The cingulum and marginal ridges are also seen from this aspect. 	The incisal aspect of lower canine is similar to that of upper except :- -The cusp tip is lingually and mesially situated (lingually due to : this teeth is lower tooth so it has lingual inclination. Mesially due to : the DCS is > MCS) -The lingual ridge is restricted to the incisal third. 

5)Pulp cavity	Upper Canine	Lower Canine
	Mesiodistally: -The pulp cavity is much narrower than in the incisors. -The pulp chamber is pointed incisally with prominent pulp horn corresponding to the cusp. -The root canal is narrow along its entire length. Labiolingually : Wider root canal and pulp chamber with pointed pulp horn .	-Similar to that of lower central incisor except. - The possibility of presence 2 seprate root canals (1 labial and 1 lingual) , if the lower canine has 2 roots .